# ENERGY GLADIATORS - HACKATHON STRATEGY GUIDE

*FOR YOUR EYES ONLY - DO NOT SHARE OR INCLUDE IN SUBMISSION*

## 1. COMPETITION TIMELINE

| When | What to Do |
|---|---|
| Friday Night | • Kickoff - receive 2026 dataset + GitHub repo access<br>• Clone official repo, start fresh project<br>• Upload recipe zip as REFERENCE only |
| Saturday (Full Day) | • MAIN BUILD DAY - judges expect active development<br>• Rebuild pipeline fresh (creates commit history)<br>• Use AI Assistant live for innovation demo<br>• Document everything in notebook |
| Saturday Night | • Polish notebook & slides<br>• Transfer to official templates<br>• Practice presentation |
| Sunday AM | • Final push to GitHub by NOON<br>• Triple-check all files named correctly<br>• Practice presentation timing (~10 min) |
| Sunday PM | • Presentations to judges<br>• Be ready for Q&A |

## 2. STEP-BY-STEP EXECUTION PLAN

### *Phase 1: Setup (Friday Night, ~30 min)*

- Clone official hackathon GitHub repo (NOT your personal repo)
- Upload recipe zip as a REFERENCE - DO NOT copy-paste code
- Start a NEW app.py - rebuild the structure from scratch
- This creates commit history showing genuine development

### *Phase 2: Data Exploration (Friday Night → Saturday AM)*

- Upload NEW 2026 training/test data
- Use AI Assistant: 'What patterns do you see in this data?'
- Document findings in notebook with markdown
- This addresses the EDA gap flagged in compliance review

### *Phase 3: Build & Adapt (Saturday)*

- Rebuild pipeline using recipe as guide
- Ask AI Assistant: 'What features should I engineer?'
- Ask AI Assistant: 'Would a different model work better?'
- DOCUMENT these AI conversations - show judges you're exploring

### *Phase 4: Polish (Saturday Night)*

- ■ Transfer work to official Jupyter notebook template
- ■ Rename to EnergyGladiators.ipynb
- ■ Add markdown explanations at EVERY step
- ■ Create slides using official PowerPoint template
- ■ ALL team member names on slide 1 (REQUIRED)

### *Phase 5: Submission (Sunday AM)*

- ■ Generate solution.csv with 100 realizations
- ■ VERIFY: Columns are Real_1 through Real_100 (not R_1)
- ■ Push everything to official GitHub repo
- ■ Final commit BEFORE noon deadline
- ■ Push 30 minutes early for safety buffer

## 3. JUDGING CRITERIA - YOUR STRATEGY

| Criterion | Weight | Your Strategy |
|---|---|---|
| Technical Accuracy | HIGH | Segmented models per fuel type (proven to work) |
| Reproducibility | 20 pts | random_state=42 everywhere, clean notebook, requirements.txt |
| Code Quality | 10 pts | Well-commented code, follow official template structure |
| Innovation | 15 pts | AI ASSISTANT - live, adaptive, industry-agnostic ML helper |
| Interpretability | 15 pts | Feature importance plots, domain-driven explanations |
| Presentation | HIGH | Live demo of AI Assistant answering questions |

## 4. YOUR KILLER DIFFERENTIATOR

**Most teams will build a model. You're building an INTELLIGENT SYSTEM that can:**

✓ Explain itself in natural language
✓ Adapt to ANY dataset the user throws at it
✓ Suggest improvements on the fly
✓ Help non-ML people understand ML

**In your presentation, say:**

*"We didn't just build a model for the 2026 hackathon problem. We built a platform that can solve FUTURE problems. Ask it anything..."*

Then do a LIVE DEMO where you type a question and it responds.

## 5. CRITICAL COMPLIANCE CHECKLIST

| Requirement | Status | Action |
|---|---|---|
| Develop during hackathon | ✓ | Rebuild fresh using recipe as guide |
| Use only 2026 official data | ✓ | Train on new data only |
| solution.csv format | ■■ VERIFY | Columns = Real_1 to Real_100 (not R_1) |
| Notebook = TeamName.ipynb | ✓ | Rename to EnergyGladiators.ipynb |
| Slides = TeamName.pptx | ✓ | All members on slide 1 |
| Push to official GitHub | ✓ | Use THEIR repo, not yours |
| 100 realizations | ✓ | Already implemented |
| Deadline: Noon Sunday | ✓ | Push 30 min early for safety |

## 6. FINAL REMINDERS

**DO NOT:**

✗ Copy-paste code from recipe (looks like pre-work)
✗ Use 2025 data for training (only 2026 official data)
✗ Forget team member names on slide 1
✗ Push to personal repo instead of official hackathon repo
✗ Include this PDF in any submission

**DO:**

✓ Rebuild pipeline fresh - use recipe as reference only
✓ Use AI Assistant LIVE during hackathon (innovation demo)
✓ Document your thought process in notebook markdown
✓ Create commit history showing genuine development
✓ Practice your presentation (aim for ~10 minutes)

## 7. RECIPE ZIP CONTENTS (Reference)

| File | Purpose |
| --- | --- |
| app.py | Complete Streamlit app with 9-step workflow + AI Assistant |
| notebooks/EnergyGladiators.ipynb | Jupyter notebook template |
| presentations/EnergyGladiators.md | Slide outline in markdown |
| requirements.txt | Python dependencies |
| data/*.csv | 2025 sample data (for structure reference only) |
| docs/HACKATHON_CHECKLIST.md | Submission checklist |

*Team Energy Gladiators - Good luck!* ■